Operations Research
2017/2018

Exercises Sheet 4
Transportation problems

1. A company that imports bananas from Madeira has three warehouses (designated A1, A2 and A3), which temporarily house imported bananas, and four stores across the country (designated L1, L2, L3 and L4), where bananas are offered for sale. It is known that A1, A2 and A3 have 9, 10 and 6 containers such bananas, respectively, per week. On the other hand, it is known that the weekly requirements of the stores L1, L2, L3 and L4 are 4, 8, 5 and 8 containers of the same, respectively. The transportation costs of each container of bananas among the several stores and the various stores of the company are also known. Such costs (tens of euros per container) are presented in the following table:

	L1	L2	L3	L4
A1	4	4	1	6
A2	9	6	2	9
A3	1	7	1	7

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *the method::*

- *of Northwest Corner;*
- *of Matrix Minimum;*
- *of Vogel's Approximation;*

c) Starting from one of the solutions obtained in b), determine the optimal solution by the *Transportation method*.

2. Suppose that it is necessary to transport a product from three origins (O1, O2 and O3) to three destinations (D1, D2 and D3). The origins O1, O2 and O3, have 6, 8, and 4 units of this product, respectively. Moreover, in D1, D2 and D3, 3, 6 and 9 units of that product are, respectively, required. The unitary transportation costs of such a product from the various origins to the various destinations are given by the following table.

	D1	D2	D3

O1	3	2	6
O2	1	7	2
O3	4	1	5

<u>Note</u>: The costs above are expressed in Monetary Units (MU)

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *the method*::

- *of Northwest Corner;*
- *of Matrix Minimum;*
- *of Vogel's Approximation;*

c) Starting from one of the solutions obtained in b), determine the optimal solution by the *Transportation method*.

3. A company has 4 factories which produce a given product and two distribution centers from where such product is distributed through the northern and southern zones of the country. In F1, F2, F3 and F4 factories are monthly produced 3, 6, 2 and 5 tons of product, respectively. The same product is then transported to two distribution centers C1 and C2, and the unitary transportation cost are given by the following table (in thousands of euros per ton):

	C1	**C2**
F1	2	4
F2	4	6
F3	1	2
F4	4	8

It is known that in centres C1 and C2, they are required 7 and 9 tons of product, respectively, per month.

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *Vogel's Approximation method;*

c) Starting from the solution obtained in b), determine the optimal solution by the *Transportation method*.

4. Suppose that it is necessary to transport a given product form three origins (O1, O2, O3) to four destinations (D1, D2, D3, D4). The origins O1, O2 and O3 have 5, 5 and 6 loadings of the product, respectively. In D1, D2, D3 and D4 they are required 4, 4, 6 and 2 loadings of the same product, respectively. Transportation costs from the various origins to the various destinations, for each shipment of the product, are given in the following table:

	D1	D2	D3	D4
O1	4	1	4	9
O2	2	6	6	0
O3	1	9	2	10

(Values in thousands of euros)

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *the Matrix Minimum method;*

c) Starting from the solution obtained in b), determine the optimal solution by the *Transportation method*.

5. Suppose that it is necessary to transport a given product from three origins (O1, O2, O3) to three destinations (D1, D2, D3). The origins O1, O2 and O3 have 7, 6 and 2 units of the product, respectively. In D1, D2 and D3 they are required 8, 4 and 3 units of the same product, respectively.

The unitary transportation costs (in euros) are given in the following table:

	D1	D2	D3
O1	6	6	7
O2	9	5	5
O3	1	2	1

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *Vogel's Approximation method.*

c) Starting from the solution obtained in b), determine the optimal solution by the *Transportation method*.

6. Now suppose that it is necessary to transport a given product from three origins (O1, O2, O3) to four destinations (D1, D2, D3, D4). Each of the origins O1, O2 and O3 has 20 units of this product. In D1, D2, D3 and D4 they are required 15, 7, 10 and 14 units of the same product, respectively. Transport costs form the various origins to the various destinations, for each product unit, are given in the following table:

	D1	**D2**	**D3**	**D4**
O1	8	2	3	5
O2	5	2	4	3
O3	9	12	5	1

(Values in thousands of euros)

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *Vogel's Approximation method.*

c) Starting from one of the solutions obtained in b), determine the optimal solution by the *Transportation method*.

7. A certain oil company has 4 production platforms and two refineries. In the platforms P1, P2, P3 and P4 there are available 3, 6, 2 and 5 oil tanks, respectively. The oil is transported by sea to the two refineries, and the unitary transportations cost are given by the following table (in thousands of euros per tank):

	R1	**R2**
P1	2	4
P2	4	6
P3	1	2
P4	4	8

It is known that in refiners R1 and R2, they are required 4 and 9 oil tanks, respectively.

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *Vogel's Approximation method;*

c) Starting from one of the solutions obtained in b), determine the optimal solution by the *Transportation method*.

8. A given company intends to transport a product from two origins (O1 and O2) to three destinations (D1, D2 and D3). The origins have 3 to 7 units of product, respectively. In destinations they are required 6, 3 and 3 units of product, respectively. The unitary transportation costs are given in the following table

	D1	**D2**	**D3**
O1	4	1	7
O2	6	3	2

Note: The costs above are expressed in Monetary Units (MU)

a) *Formulate the problem* in terms of a linear programming model in order to minimize the total transportation cost;

b) Determine an initial feasible basic solution for the problem using *the Northwest Corner method;*

c) Starting from one of the solutions obtained in b), determine the optimal solution by the *Transportation method*.